GCSE BIOLOGY

Higher Tier – Paper 2 style

Topic 4.7: Ecology

Practice questions in the style of AQA GCSE Biology (8461). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 210 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **198**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 198

0 1 Ecology is the study of how organisms interact with each other and their environment.

0 1 . 1 Draw **one** line from each term to its correct meaning. **[4 marks]**

Term	Meaning
population	All the living things and the non-living parts in an area
community	The place where an organism lives
habitat	All the organisms of one species in an area
ecosystem	All the populations of different species in an area

0 1 . 2 Give **two** abiotic factors that can affect a community. **[2 marks]**

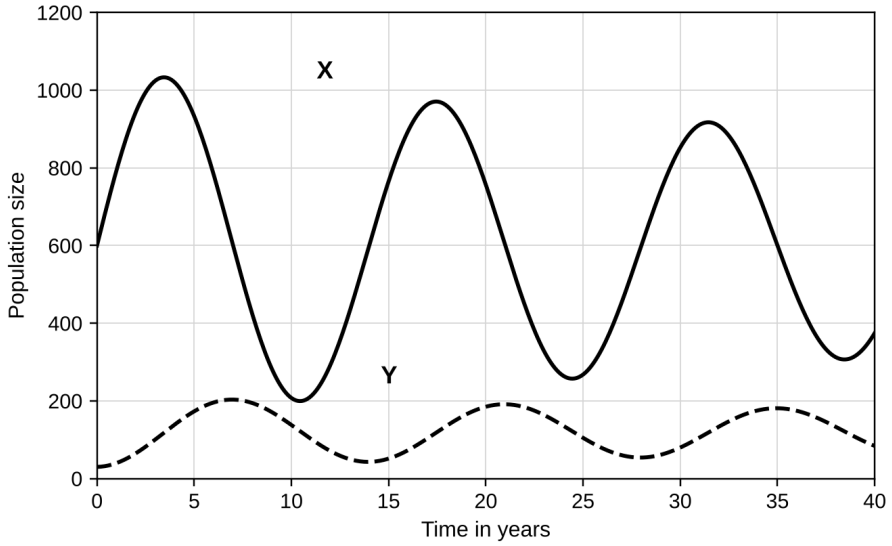
0 1 . 3 Give **two** biotic factors that can affect a community. **[2 marks]**

0 1 . 4 Animals compete for resources. Name **two** things that animals compete for. **[2 marks]**

0 1 . 5 Plants compete for resources. Name **two** things that plants compete for. **[2 marks]**

0 1 . 6 Explain what is meant by **interdependence**. **[2 marks]**

0 2 The graph shows how the population sizes of two species, **X** and **Y**, in a stable community changed over 40 years. One species is a predator and one is its prey.



0 2 . 1 Which species, **X** or **Y**, is the prey? **[1 mark]**

0 2 . 2 Give the evidence from the graph for your answer to 2.1. **[1 mark]**

0 2 . 3 Explain why the number of Y rises after the number of X rises. **[2 marks]**

0 2 . 4 Explain why the number of X falls after the number of Y has risen. **[2 marks]**

0 2 . 5 Use the graph to estimate the time between the first peak of X and the first peak of Y. **[2 marks]**

Answer = _____ years

0 2 . 6 Describe **one** change that could cause a long-term fall in the number of Y without affecting X. **[2 marks]**

0 3 Organisms are adapted to survive in their habitats.

0 3 . 1 What is meant by an **adaptation**? **[1 mark]**

0 3 . 2 Name the **three** types of adaptation. **[3 marks]**

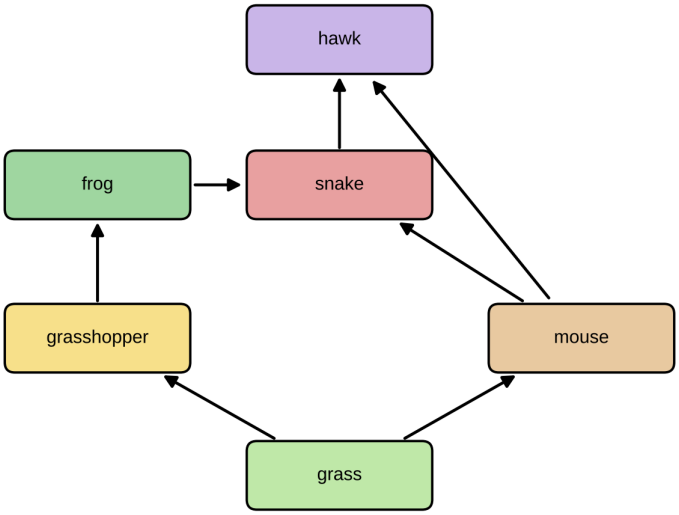
0 3 . 3 A polar bear lives in the Arctic. It has thick fur, a thick layer of fat and white fur. Explain how **two** of these adaptations help the polar bear survive. **[4 marks]**

0 3 . 4 A cactus lives in a desert. Its stem is thick and stores water and its leaves are spines. Explain how these features help the cactus survive. **[3 marks]**

0 3 . 5 Some organisms are **extremophiles**. What does this mean? **[1 mark]**

0 3 . 6 Bacteria live in deep-sea hydrothermal vents where the temperature is over 80 °C. Suggest how these bacteria are adapted to survive. **[2 marks]**

0 4 The diagram shows a food web in a grassland ecosystem.



0 4 . 1 Name the producer in the food web. **[1 mark]**

0 4 . 2 Write down a food chain from the web that has **four** organisms. **[2 marks]**

0 4 . 3 Name an organism that is a secondary consumer. **[1 mark]**

0 4 . 4 Name the **apex predator** in this food web. **[1 mark]**

0 4 . 5 Disease kills most of the mice. Predict the effect on the numbers of the snake and the grasshopper. Explain your answers. **[4 marks]**

0 4 . 6 Why do the arrows in a food web point from the organism being eaten to the organism that eats it? **[1 mark]**

0 5 A student investigated the number of daisy plants in a school field. The field is a rectangle 40 m by 10 m. She used a 1 m × 1 m quadrat. The table shows her results.

Quadrat	**1**	**2**	**3**	**4**	**5**
Number of daisy plants	4	7	3	6	5

0 5 . 1 Describe how the student should place the quadrat to get a **representative** sample. **[3 marks]**

0 5 . 2 Calculate the mean number of daisy plants per m². **[2 marks]**

Answer = _____ per m²

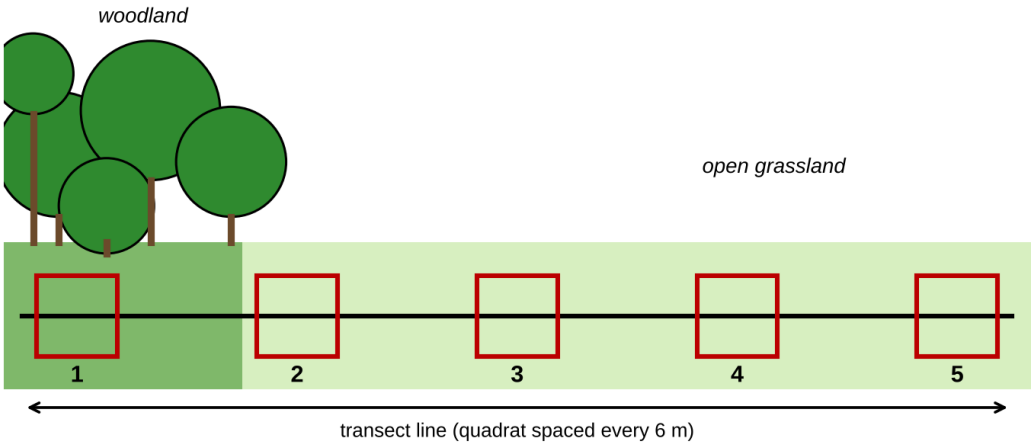
0 5 . 3 Estimate the total number of daisy plants in the whole field. **[2 marks]**

Answer = _____ plants

0 5 . 4 Suggest **two** ways the student could make her estimate more accurate. **[2 marks]**

0 5 . 5 Another student found 5 daisies per m² at the edge of the field and 0.5 per m² near the trees. Suggest **one** abiotic factor that could explain this. **[1 mark]**

0 6 A scientist used a transect to investigate how the number of bluebells changes from a woodland into an open grassland. The diagram shows the five positions where a 1 m² quadrat was placed. The table shows the results.



Quadrat	1	2	3	4	5
Bluebells per m²	24	16	7	2	0

0 6 . 1 Describe how the scientist should have set up the transect. **[3 marks]**

0 6 . 2 Calculate the mean number of bluebells per m² in the **woodland** quadrats 1 and 2. **[2 marks]**

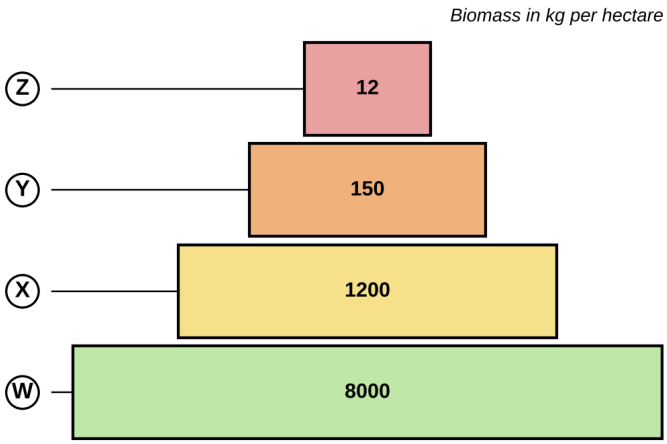
Answer = _____ per m²

0 6 . 3 Describe the trend shown in the results. **[2 marks]**

0 6 . 4 Suggest **one** abiotic factor that could explain the trend. **[1 mark]**

0 6 . 5 Explain why a transect is better than random sampling for this investigation. **[2 marks]**

0 7 The diagram shows a pyramid of biomass for a food chain in a field. The letters **W** to **Z** show the trophic levels.



0 7 . 1 Which trophic level (**W**, **X**, **Y** or **Z**) contains the producers? **[1 mark]**

0 7 . 2 Which letter represents the tertiary consumers? **[1 mark]**

0 7 . 3 Calculate the percentage of biomass transferred from **W** to **X**. **[2 marks]**

Answer = _____ %

0 7 . 4 Calculate the percentage of biomass transferred from **X** to **Y**. **[2 marks]**

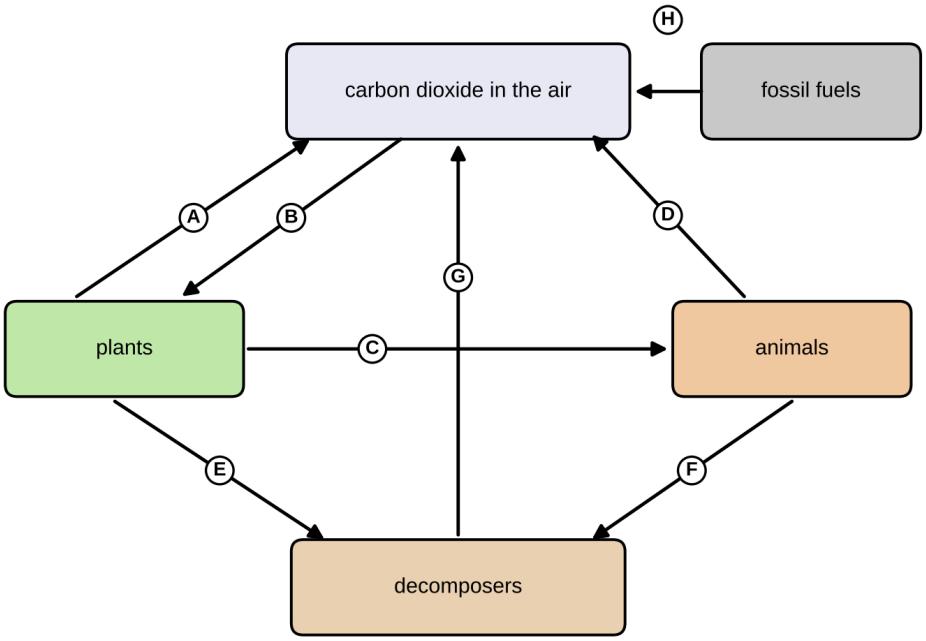
Answer = _____ %

0 7 . 5 Give **three** reasons why not all the biomass is transferred to the next trophic level. **[3 marks]**

0 7 . 6 Explain why food chains rarely have more than five trophic levels. **[2 marks]**

0 7 . 7 Explain why a pyramid of biomass is always a pyramid shape, but a pyramid of numbers may not be. **[2 marks]**

0 8 The diagram shows part of the carbon cycle. The letters **A** to **H** show processes.



0 8 . 1 Name process **B**. **[1 mark]**

0 8 . 2 Name the process shown by arrows **A**, **D** and **G**. **[1 mark]**

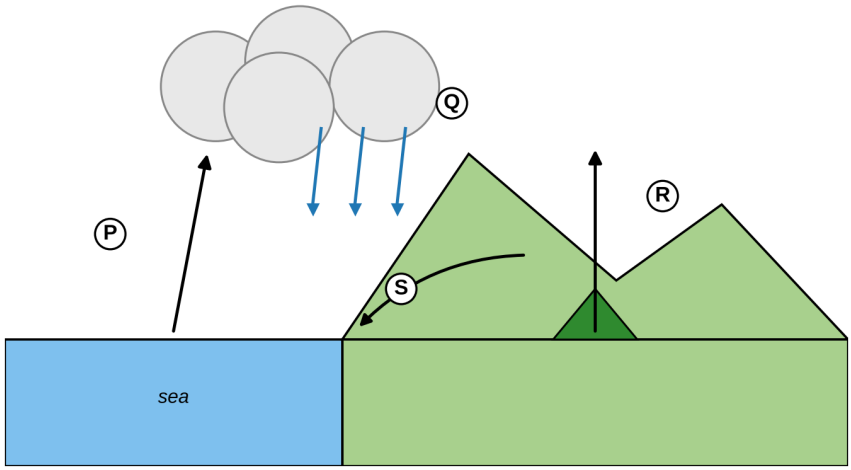
0 8 . 3 Name the process shown by arrow **H**. **[1 mark]**

0 8 . 4 What does arrow **C** represent? **[1 mark]**

0 8 . 5 Explain what happens in arrows **E** and **F**. **[2 marks]**

0 8 . 6 Explain why the amount of carbon dioxide in the atmosphere has been increasing for the last 200 years. **[3 marks]**

0 9 The diagram shows the water cycle.



0 9 . 1 Name the processes **P**, **Q**, **R** and **S**. **[4 marks]**

0 9 . 2 Describe what happens inside the cloud before rain falls. **[2 marks]**

0 9 . 3 Explain why the Sun is needed for the water cycle. **[2 marks]**

0 9 . 4 Fresh water is needed for drinking. Describe the stages of making potable water from fresh water. **[3 marks]**

1 0 A gardener makes compost. Dead plant material is mixed with soil and left in a heap.

1 0 . 1 Name the organisms that break down dead plant material. **[1 mark]**

1 0 . 2 Give **three** conditions that increase the rate of decay. **[3 marks]**

1 0 . 3 Explain why the rate of decay is slower in a very cold compost heap. **[2 marks]**

1 0 . 4 Explain why a waterlogged heap decays slowly at first. **[2 marks]**

10.5 Suggest why farmers add compost to the soil. **[2 marks]**

10.6 Some farms use anaerobic decay in a biogas generator. Describe how this helps. **[2 marks]**

11 Human activities affect biodiversity.

11.1 What is meant by **biodiversity**? **[1 mark]**

11.2 Why is it important to maintain high biodiversity? **[2 marks]**

11.3 Give **two** reasons why the Earth's human population is growing at an increasing rate. **[2 marks]**

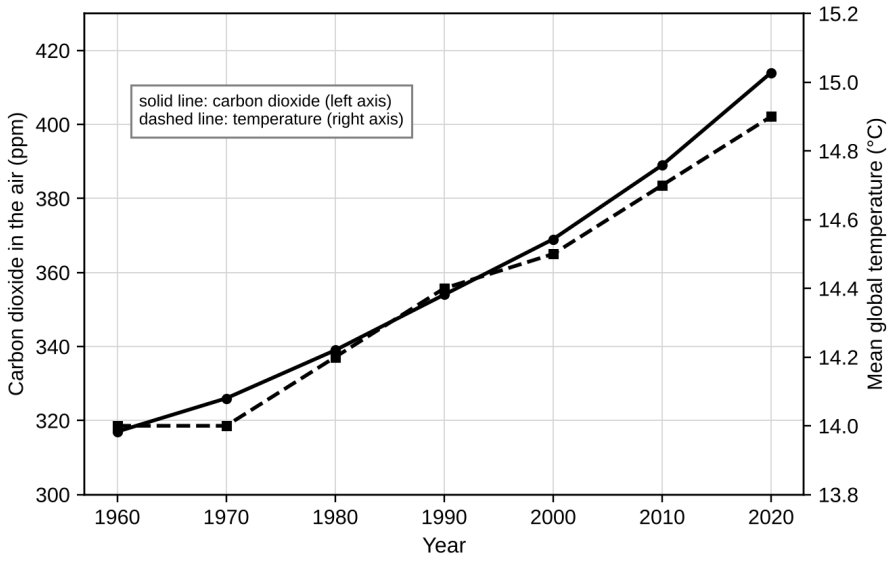
11.4 Give **three** ways that humans reduce the amount of land available for other organisms. **[3 marks]**

1 1 . 5 Describe how **one** human activity can pollute the water. **[2 marks]**

1 1 . 6 Describe how **one** human activity can pollute the air. **[2 marks]**

1 1 . 7 Peat bogs are being destroyed to produce garden compost. Explain why this is a problem. **[3 marks]**

1 2 The graph shows the concentration of carbon dioxide in the air and the mean global temperature from 1960 to 2020.



1 2 . 1 Calculate the percentage increase in carbon dioxide from 1960 (318 ppm) to 2020 (414 ppm). **[3 marks]**

Answer = _____ %

1 2 . 2 Describe the relationship between the carbon dioxide concentration and the temperature. **[2 marks]**

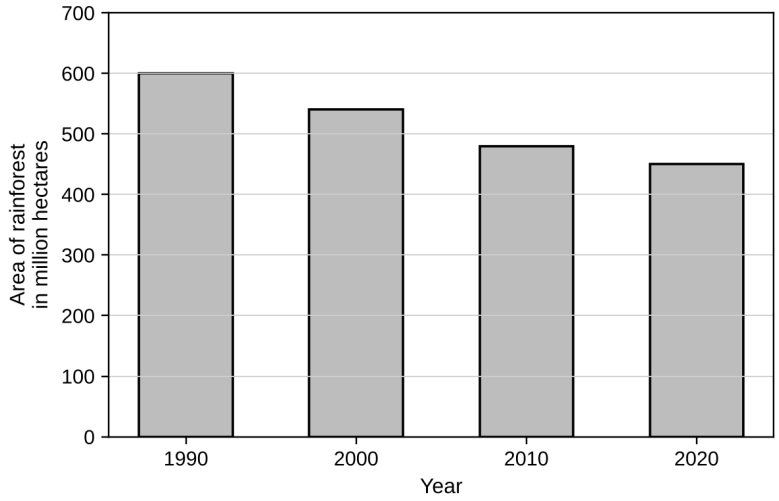
1 2 . 3 A student says: 'The graph proves that carbon dioxide causes global warming.' Evaluate the statement. **[2 marks]**

1 2 . 4 Name **two** greenhouse gases. **[2 marks]**

1 2 . 5 Explain how greenhouse gases cause the Earth to warm up. **[2 marks]**

1 2 . 6 Describe **three** possible consequences of global warming. **[3 marks]**

1 3 Large areas of tropical rainforest are being cleared. The bar chart shows the area of rainforest in one country.



1 3 . 1 Calculate the percentage decrease in the area of rainforest from 1990 to 2020. **[3 marks]**

Answer = _____ %

1 3 . 2 Calculate the mean area lost per year between 1990 and 2020. **[2 marks]**

Answer = _____ million hectares

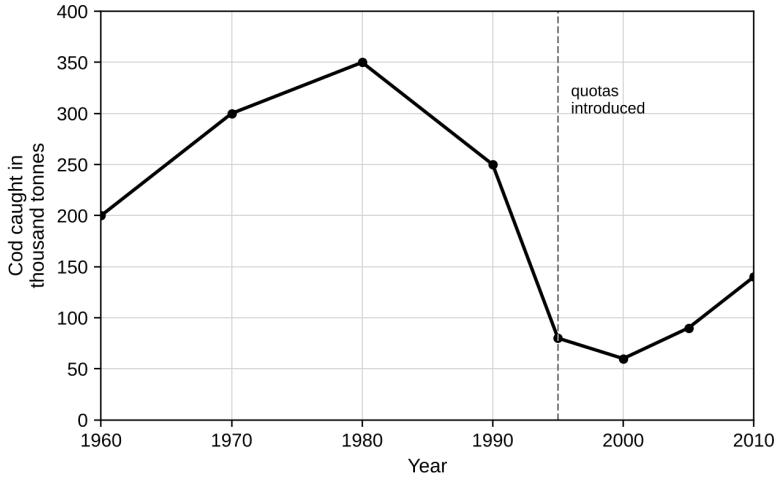
1 3 . 3 Give **two** reasons why large areas of rainforest are cleared. **[2 marks]**

1 3 . 4 Explain how deforestation increases the amount of carbon dioxide in the air. **[3 marks]**

1 3 . 5 Describe **three** programmes that conserve biodiversity. **[3 marks]**

1 3 . 6 Suggest why some local people may not support conservation programmes. **[2 marks]**

1 4 The graph shows the mass of cod caught in a sea from 1960 to 2010. International quotas on cod fishing were introduced in 1995.



1 4 . 1 State the year of the greatest mass of cod caught. **[1 mark]**

1 4 . 2 Calculate the percentage decrease from the maximum catch to the catch in 2000. **[3 marks]**

Answer = _____ %

1 4 . 3 Explain why catches fell sharply in the years before 1995. **[2 marks]**

1 4 . 4 Explain how quotas help to maintain fish stocks. **[3 marks]**

1 4 . 5 Use the graph to evaluate whether the quotas have worked. **[2 marks]**

1 4 . 6 **Food security** is having enough food to feed a population. Give **two** factors that threaten food security. **[2 marks]**

1 5 Farmers use different methods to produce more food.

1 5 . 1 Explain how **one** method of reducing the amount of energy lost in a food chain can increase food production in farm animals. **[3 marks]**

1 5 . 2 Give **one** ethical objection to intensive farming of animals. **[1 mark]**

1 5 . 3 Mycoprotein is made from a fungus grown in fermenters. Suggest **two** advantages of mycoprotein as a food. **[2 marks]**

1 5 . 4 Describe how a GM crop can help to feed a growing population. **[2 marks]**

1 5 . 5 Give **two** concerns about the use of GM crops. **[2 marks]**

1 6 Stretch questions (grade 7–9).

1 6 . 1 *Describe how carbon moves through a woodland ecosystem and explain how the carbon cycle is affected when a large area of the woodland is burned. **[6 marks]**

1 6 . 2 *A lake receives fertiliser run-off from nearby farms. Explain how the fertiliser can cause the death of fish in the lake. **[6 marks]**

1 6 . 3 *Evaluate the idea that farmers should stop using intensive methods and return to traditional methods to feed the world population. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.7: Ecology

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	population = all the organisms of one species in an area ; community = all the populations of different species living in an area ; habitat = the place where an organism lives ; ecosystem = the community plus the non-living (abiotic) parts of the environment ;	Award one mark for each correct match. Allow the meanings written next to the correct term.	4	AO1 4.7.1.1
01.2	any two from: light intensity ; temperature ; moisture level ; soil pH / mineral content ; wind intensity and direction ; carbon dioxide level (plants) ; oxygen level (aquatic) ;	Do not accept 'weather' alone.	2	AO1 4.7.1.2
01.3	any two from: availability of food ; new predators arriving ; new pathogens ; one species outcompeting another ;	Allow competition for mates / pollinators.	2	AO1 4.7.1.2
01.4	any two from: food ; water ; mates ; territory / space ;		2	AO1 4.7.1.1
01.5	any two from: light ; water ; space ; mineral ions / nutrients (from the soil) ;		2	AO1 4.7.1.1
01.6	species depend on other species in the community ; for food, shelter, pollination or seed dispersal / a change in one species can affect others ;		2	AO1 4.7.1.1
02.1	X		1	AO3 4.7.1.1 / WS 3.5
02.2	prey population is larger / peaks first, then predator rises after ;	Allow: prey number rises first and the predator number follows (lags) behind.	1	AO3 WS 3.5
02.3	more food (prey) available for the predator ; so more predators survive and reproduce ;		2	AO2 4.7.1.1
02.4	more predators eat more prey ; so prey numbers fall ;		2	AO2 4.7.1.1
02.5	X peak at about 3 to 4 years and Y peak at about 7 years ; about 3 to 4 years ;		2	AO3 WS 3.1
02.6	any one of: a new disease affects the predator ; a new predator / competitor of Y arrives ; habitat of Y is destroyed ; hunting / pollution kills Y ; linked effect described (e.g. fewer Y survive to breed) ;		2	AO3 4.7.1.2
03.1	a feature that helps an organism survive in its environment / habitat		1	AO1 4.7.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	structural ; behavioural ; functional (physiological) ;		3	AO1 4.7.1.3
03.3	thick fur / fat: reduces heat loss / insulation ; keeps the bear warm in very cold conditions ; white fur: camouflage in snow ; so the bear can sneak up on prey / is not seen by prey ;	Allow a small surface area to volume ratio (large body size) as an alternative adaptation.	4	AO2 4.7.1.3
03.4	stem stores water for dry periods ; spines have a small surface area so less water is lost by transpiration ; spines protect against herbivores / deep or wide root systems to reach water ;	Any three.	3	AO2 4.7.1.3
03.5	they live in extreme conditions (e.g. high temperature, pressure, salt concentration)		1	AO1 4.7.1.3
03.6	their enzymes do not denature at high temperature ; they get energy from chemicals (e.g. chemosynthesis) not light ;	Allow: any suitable adaptation to high pressure.	2	AO3 4.7.1.3
04.1	grass		1	AO2 4.7.2.1
04.2	any correct four-organism chain, e.g. grass → grasshopper → frog → snake ; arrows in the correct direction ;	Also allowed: grass → mouse → snake → hawk; grass → grasshopper → frog → snake.	2	AO2 4.7.2.1
04.3	frog / snake (in the chain with grasshopper / mouse) ;	Allow: snake or hawk when eating the mouse. Do not accept grasshopper or mouse.	1	AO2 4.7.2.1
04.4	hawk		1	AO2 4.7.2.1
04.5	snake numbers decrease ; less food for the snake (but it can still eat frogs) ; grasshopper numbers increase ; less competition for the grass (mouse numbers fall) ;	Allow a reasoned answer that snakes decrease only a little because they also eat frogs.	4	AO3 4.7.2.1
04.6	the arrow shows the direction of energy / biomass transfer ;		1	AO1 4.7.2.1
05.1	use random positions ; e.g. use random numbers as coordinates on a grid / tape measures ; avoid bias / do not choose where the daisies are ;	Allow throwing the quadrat over the shoulder only if bias is removed.	3	AO2 4.7.1.4 / RP
05.2	(4 + 7 + 3 + 6 + 5) ÷ 5 ; 5 (per m^2) ;		2	AO2 MS 2b
05.3	area = 40 × 10 = 400 m^2 ; 400 × 5 = 2000 ;		2	AO2 MS 2b
05.4	use more quadrats / larger sample ; use bigger / more quadrats positioned randomly ; repeat on different days ;	Do not accept 'repeat the experiment' without detail. Allow: count the field in sections.	2	AO3 4.7.1.4 / RP
05.5	less light under trees / drier soil / lower soil mineral content ;		1	AO3 4.7.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	lay a tape measure / line from the woodland into the grassland ; place the quadrat at regular intervals (every 6 m) ; count the bluebells in each quadrat ;		3	AO1 4.7.1.4 / RP
06.2	(24 + 16) ÷ 2 ; 20 per m^2 ;		2	AO2 MS 2b
06.3	number of bluebells decreases ; as distance from the woodland increases / moves into grassland ;		2	AO3 WS 3.5
06.4	light intensity (bluebells grow in spring before trees are in leaf) / soil moisture / temperature ;		1	AO3 4.7.1.2
06.5	a transect shows change along a gradient / across a distance ; random sampling would give a single average and miss the change ;		2	AO3 4.7.1.4
07.1	W		1	AO2 4.7.2.2
07.2	Z		1	AO2 4.7.2.2
07.3	1200 ÷ 8000 × 100 ; 15 % ;		2	AO2 MS 1c
07.4	150 ÷ 1200 × 100 ; 12.5 % ;		2	AO2 MS 1c
07.5	some biomass is lost as waste (faeces, urine) ; some is not eaten (e.g. bones, roots, fur) ; some is used in respiration (released as heat / carbon dioxide) ;	Allow: movement uses energy; not all parts digestible. Any three.	3	AO1 4.7.2.2
07.6	energy / biomass lost at each stage ; so not enough energy remains to support more organisms ;		2	AO2 4.7.2.2
07.7	biomass always decreases up the food chain (energy lost) ; numbers can be inverted if one large producer (e.g. tree) supports many small consumers ;		2	AO3 4.7.2.2
08.1	photosynthesis		1	AO1 4.7.3.2
08.2	respiration		1	AO1 4.7.3.2
08.3	combustion (burning)		1	AO1 4.7.3.2
08.4	feeding / eating (of plants by animals)		1	AO1 4.7.3.2
08.5	dead plants and animals / waste (carbon compounds) pass to decomposers ; decomposers break them down ;		2	AO2 4.7.3.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.6	burning fossil fuels releases carbon dioxide ; deforestation / fewer trees remove less carbon dioxide by photosynthesis ; rate of release is greater than the rate of removal ;		3	AO2 4.7.3.2 / 4.7.3.5
09.1	P: evaporation ; Q: precipitation (rain) ; R: transpiration (evaporation from plants) ; S: run-off (river flow / surface water flowing to the sea) ;		4	AO1 4.7.3.1
09.2	water vapour cools ; condenses into droplets ;		2	AO1 4.7.3.1
09.3	energy from the Sun heats the water ; causing evaporation (and transpiration) ;		2	AO2 4.7.3.1
09.4	choose a suitable source ; filter to remove solid material ; sterilise with chlorine / ozone / ultraviolet ;		3	AO1 4.7.3.1
10.1	decomposers (bacteria and fungi)		1	AO1 4.7.3.2
10.2	warm temperature ; moisture / water ; oxygen (air) ; cut into small pieces ;	Any three.	3	AO1 4.7.3.2
10.3	decomposers have less kinetic energy / enzymes work slowly ; fewer collisions between enzymes and substrate ;		2	AO2 4.7.3.2
10.4	little oxygen present ; so aerobic microorganisms cannot respire fast ;		2	AO2 4.7.3.2
10.5	decomposers release minerals / nitrates into the soil ; for plants to absorb for healthy growth ;		2	AO3 4.7.3.2
10.6	microorganisms break down waste without oxygen producing methane (biogas) ; used as a fuel for cooking / heating / electricity ;		2	AO1 4.7.3.2
11.1	the variety of all the different species of organisms on Earth / in an ecosystem		1	AO1 4.7.3.3
11.2	species depend on each other so an ecosystem is more stable ; provides food, medicine and raw materials / maintains the supply of resources ;		2	AO1 4.7.3.3
11.3	any two: medicine / lower death rates ; improved food production ; better sanitation / disease control ;		2	AO1 4.7.3.3
11.4	building / housing ; quarrying ; farming ; dumping waste ;	Any three.	3	AO1 4.7.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
11.5	sewage / fertiliser run-off / toxic chemicals from industry ; linked effect: eutrophication or kills organisms ;		2	AO1 4.7.3.3
11.6	smoke / acidic gases (sulfur dioxide) from burning fuels ; acid rain / respiratory problems ;		2	AO1 4.7.3.3
11.7	habitats of organisms are lost / reduced biodiversity ; peat decays or is burned releasing carbon dioxide ; carbon dioxide adds to global warming ;		3	AO2 4.7.3.4
12.1	414 − 318 = 96 ; 96 ÷ 318 × 100 ; 30.2 (%) ;		3	AO2 MS 1c
12.2	both increase ; positive correlation ;		2	AO3 WS 3.5
12.3	correlation does not prove cause ; other factors (e.g. methane, Sun) could also affect temperature ; scientific evidence for the mechanism supports the link ;		2	AO3 4.7.3.5
12.4	carbon dioxide ; methane (water vapour) ;		2	AO1 4.7.3.5
12.5	they absorb long-wavelength / infrared radiation emitted from the Earth ; and re-emit some back to the surface ;		2	AO1 4.7.3.5
12.6	sea level rises (ice melts) ; extreme weather / changes in rainfall ; changes in species distribution / habitat loss / migration ;	Any three. Allow: changes in crop yields.	3	AO1 4.7.3.5
13.1	600 − 450 = 150 ; 150 ÷ 600 × 100 ; 25 (%) ;		3	AO2 MS 1c
13.2	150 ÷ 30 ; 5 million hectares per year ;		2	AO2 MS 1a
13.3	cattle and rice fields ; grow crops for biofuels / palm oil ;	Allow: logging, building / mining.	2	AO1 4.7.3.4
13.4	fewer trees to take in carbon dioxide ; less photosynthesis ; burning / decay of felled trees releases carbon dioxide ;		3	AO2 4.7.3.4
13.5	breeding programmes for endangered species ; protection and regeneration of rare habitats ; reintroduction of field margins / hedgerows ; reduction of deforestation and carbon dioxide emissions ; recycling rather than landfill ;	Any three.	3	AO1 4.7.3.6
13.6	they need the land or jobs / income from farming or logging ; cost of programmes / conflict with local needs ;		2	AO3 4.7.3.6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
14.1	1980		1	AO3 WS 3.1
14.2	350 − 60 = 290 ; 290 ÷ 350 × 100 ; 82.9 (%) (allow 83) ;		3	AO2 MS 1c
14.3	overfishing / too many fish caught ; too few adults left to breed ;		2	AO3 4.7.3.7
14.4	limit the number / mass of fish that can be caught ; enough fish left to breed ; and larger mesh nets (allow fish to escape) so young fish survive ;	Allow: fishing bans in breeding season.	3	AO2 4.7.3.7
14.5	catch increased from 2000 (60) to 2010 (140) ; but still below the 1980 level so recovery is incomplete ;		2	AO3 WS 3.5
14.6	any two: increasing birth rate ; changing diets ; new pests / pathogens ; environmental changes (e.g. drought) ; cost of agricultural inputs ; conflicts ;		2	AO1 4.7.3.7
15.1	restrict animal movement / keep indoors ; less energy lost in movement / by respiration ; keep warm so less energy used to maintain body temperature ;		3	AO2 4.7.3.7
15.2	animals suffer / are kept in poor conditions / have less space ;		1	AO3 4.7.3.7
15.3	high protein ; produced quickly / needs little land ; can be made from waste (glucose syrup) ;	Any two.	2	AO3 4.7.3.7
15.4	genes inserted give resistance to pests / drought, or higher yield ; so more food per hectare ;		2	AO1 4.6.2.4 / 4.7.3.7
15.5	effect on wild flowers / insects / food chains ; effects on human health unknown ; cost of GM seeds ; gene may spread to wild plants ;	Any two.	2	AO1 4.6.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
16.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • plants take in carbon dioxide by photosynthesis to make glucose / organic compounds • carbon passes to animals by feeding • carbon returned to the air by respiration of plants, animals and decomposers • dead material decayed by decomposers returning carbon dioxide • burning (combustion) releases carbon dioxide rapidly • fewer trees so less photosynthesis, so more carbon dioxide stays in the air / contributes to global warming		6	AO2 4.7.3.2
16.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • fertiliser contains nitrates and phosphates, so algae grow rapidly (algal bloom) • algae cover the surface so light cannot reach plants below • submerged plants die because they cannot photosynthesise • bacteria decompose dead plants and algae and increase in number • decomposers use up the oxygen by respiration • fish and other aquatic animals suffocate / die (eutrophication)		6	AO2 4.7.3.3 / 4.7.3.5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
16.3	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • intensive: higher yield per hectare, more food from less land, lower cost per unit • intensive: damage to environment, use of fertiliser / pesticides, reduced biodiversity, eutrophication • intensive animal farming: efficient energy transfer but ethical concerns • traditional: more biodiversity, fewer chemicals, less pollution • traditional: lower yields, may not feed the growing population, more land needed (more deforestation) • balanced conclusion linked to food security, e.g. combine methods / sustainable approaches		6	AO3 4.7.3.7

Where marks are lost in Topic 4.7

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Quadrats:** random positions (random numbers as coordinates). Mean per m² × total area = population estimate. A **transect** is for change along a gradient.
- ☐ **Percentage transfer** = biomass in the next level ÷ biomass in the level below × 100. Losses: respiration, waste, parts not eaten.
- ☐ **Food chains and webs:** arrows show the direction of energy transfer. Predict effects in steps: 'fewer prey → less food → predators fall'.
- ☐ **Predator–prey graphs:** prey peaks first; predator follows. Say **which** graph line and **why**.
- ☐ **Carbon cycle:** photosynthesis removes CO_2; respiration, combustion and decay (decomposers' respiration) release it.
- ☐ **Rate of decay:** warmth, moisture, oxygen, small pieces. Anaerobic decay makes methane (biogas).
- ☐ **Global warming:** greenhouse gases absorb infrared radiation. Correlation does not mean cause — say 'other factors could...'.
- ☐ **Eutrophication chain:** nitrates → algal bloom → light blocked → plants die → decomposers use oxygen → fish die. Learn every link.
- ☐ **Evaluate questions:** give advantages, disadvantages, then a judgement ('overall…').
- ☐ **Percentage change = change ÷ original × 100.** Show the working: the method mark is easy to gain.